

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination Spring-2026
Course Code: GEBL-2401
Course Title: Bangla Language and Literature
Full Marks: 50 Time: 2.5 Hours

ক-বিভাগ
ভাষা ও নিমিতি: ৩০
(ডান পাশের সংখ্যা প্রশ্নের মান জ্ঞাপক)

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০১.	প্রয়োগসহ বাংলা বানানের দশটি নিয়ম উদাহরণসহ লেখ।	১০	CLO1	Apply
০২.	পত্রিকায় প্রকাশের উপযোগী “ধর্ষণ প্রতিরোধে চাই দৃষ্টান্তমূলক শান্তি” - শীর্ষক একটি পত্র রচনা কর। অথবা একটি স্বরচিত ক্ষুদ্রে গল্প উপস্থাপন কর।	১০ ১০	CLO1 CLO1	Create Create
০৩.	সংক্ষেপে আলোচনা কর: ক. মানবতা ও নৈতিকতা। খ. মুক্তিযুদ্ধের পঞ্চাঙ্গ বছর: প্রাপ্তি-অপ্রাপ্তি।	৫×২=১০	CLO1	Create

খ-বিভাগ
সাহিত্য: ২০

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০৪.	“আজ সৃষ্টি সুখের উল্লাসে” কবিতায় প্রতিফলিত কবি কাজী নজরুল ইসলামের জীবনদৃষ্টির পরিচয় দাও। অথবা ‘মানুষের প্রতি বিশ্বাস হারানো পাপ’ উক্তির আলোকে রবীন্দ্রনাথ ঠাকুরের রাজনৈতিক দৃষ্টিভঙ্গি তুলে ধর।	১০ ১০	CLO3 CLO3	Evaluate Analyze
০৫.	“কবর” নাটকে প্রতিফলিত চরিত্রের মাধ্যমে রাষ্ট্রের দায়িত্বশীলদের পচনশীলতার চিত্র আলোচনা কর।	১০	CLO3	Evaluate

- (i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions			
CO	CO Statements	PO	BL
CO1	Explain the basic concept of financial accounting, cost accounting and management accounting.	PO11	C2
CO2	Analyze the basic concept of Cost Accounting and preparation of Cost Sheet.	PO11	C4
CO3	Apply the tools from accounting and cost accounting this would facilitate the decision making i.e. Budgeting, Make or Buy decision.	PO11	C3
CO4	Compare the different business situations and suggest to best solution with analytical abilities.	PO11	C5

Bloom's Levels (BL) of the Questions						
Letter Symbols	C1	C2	C3	C4	C5	C6
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Answer the following questions

Part A

1. The unadjusted trial balance of Aracom service company at July 31, 2025, and the related month-end adjustment data are as follows: CO2 C2 10

	Dr (\$)	Cr (\$)
Cash	15,000	
Accounts Receivable	13,000	
Prepaid insurance	3,600	
Supplies	1,000	
Furniture	16,800	
Accumulated Depreciation- furniture		5,000
Accounts Payable		3,450
Salary Payable		-
Aracom service Capital		38,650
Aracom service Withdrawals	14,000	
Legal Service Revenue		19,250
Salary Expense	2,400	
insurance Expense	-	
Utilities Expense	550	
Depreciation Expense-furniture		
Supplies Expense		
Total	66,350	66,350

Adjustment data:

- The unadjusted prepaid balance of insurance \$3,600 relates to the period July through September. Prepaid insurance expired during the month.
- Supplies on hand at July 31, \$500.
- Depreciation on furniture for the month. The estimated useful life of the furniture is four years.
- Accrued salary expense at July 31 for three days only. The five-day weekly payroll is \$1,000.
- Accrued legal service revenue at July 31, \$1,000.

Required:

- Prepare the adjusted trial balance.

2.

K.M. Attorney
Adjusted Trial Balance as in 31 December 2023

CO2 C3 10

Particular	Dr(\$)	Cr(\$)
Cash	90,000	
Accounts receivable	20,000	
Interest Receivable	200	
Notes Receivable	4,000	
Prepaid Insurance	960	
Prepaid Rent	2,400	
Supplies	1,520	
Equipment	Last Four digit of student ID	
Depreciation- Equipment		12,500
Buildings	140000	
Depreciation- Buildings		15,000
Land	56,240	
Accounts payable		Last Four digit of student ID
Notes Payable		10,920
Tax Payable		10,000
Interest payable		750
Salaries payable		7,000
Unearned revenue		20,200
Capital		100000
Drawings	40,000	
Service revenue		360000
Advertising expense	1200	
Salary expense	150000	
Rental expense	9,600	
Supplies expense	2,280	
Office expense	10,000	
Interest expense	750	
Insurance expense	1,920	
Dep. expense- Equipment	2,500	
Dep. expense- Building	3,000	
Interest revenue		200
Total	536,570	536,570

Required: Financial statements of K.M. Attorney for the year ended 31 December 2023.

Or,

The trial balance of Marham Healthcare at December 31, 2024 and data needed for the month-end adjustments are as follows:

CO3 C4 10

- i) Supplies on hand at the month ended, Tk. 340. ii) Prepaid rent still in force at December 31, Tk. 1,800. iii) Depreciation on furniture and fixtures for the month Tk. 500. iv) Depreciation on building for the month Tk. 800. v) Accrued salary expense at December 31, Tk. 1,200. vi) Unearned service revenue still unearned at December 31, Tk. 4,300. vii) Accrued service revenue at December 31, Tk. 1,500.

Marham Healthcare
Trial Balance
As in December 31, 2024

Sl. No.	Name of Accounts	Ref.	Debit (Tk.)	Credit (Tk.)
1	Cash		10,400	-
2	Accounts Receivable		12,310	-
3	Supplies		1,840	-
4	Prepaid Rent		2,700	-
5	Furniture		26,840	-
6	Accumulated Depreciation-Furniture		-	3,400
7	Building		68,300	-
8	Accumulated Depreciation-Building		-	9,100
9	Accounts Payable		-	7,290
10	Salary Payable		-	-
11	Unearned Service Revenue		-	5,300
12	Capital		-	85,500
13	Withdrawal		4,900	-
14	Service Revenue		-	19,560
15	Salary Expense		1,840	-
16	Rent Expense		-	-
17	Utilities Expense		530	-
18	Depreciation Expense-Furniture		-	-
19	Depreciation Expense-Building		-	-
20	Supplies Expense		-	-
21	Advertisement Expense		490	-
	Total		<u>130,150</u>	<u>130,150</u>

Required: Prepare a worksheet of the Marham Healthcare.

Part B

3. The following information has been taken from the records of Aramco company.

Marketing and Administrative expense	\$14,000
Selling agent commission	5,000
Materials Purchased ✓	\$1,50,000
Purchase Discounts ✓	800
Direct Labor	2,00,000
Purchase Returns and Allowance ✓	4,000
Freight In	2,500
Sales Salaries and Expenses	17,000
Other Factory Expenses	3,000
Freight Out	2,000
Factory Insurance	13,000
Depreciation-Machinery	35,500
Sales	

CO2 C4 10

	4,55,000	
Salary of factory supervisor	4,500	
Inventories	1-Jan	31-Dec
Materials	\$5000	\$3,000
Work In Process	\$14,000	\$9,000
Finished Goods	\$10,000	\$6,000

Requirements:

prepare a cost of goods sold statement for the year ended December 31, 2025

4. Reeven products sell camping equipment. One of the company's products-a camo lantern, sells for \$90 per unit. Variable expenses are \$63 per lantern, and fixed expenses associated with the lantern total \$135,000 per month. CO3 C3, 10
C4
- Required:
- Compute the company's break-even point in number of lanterns and in total sales dollars.
 - If the variable expenses per lantern increase as a percentage of the selling price, will it result in a higher or a lower break-even point? Why? (Assume that the fixed expenses remain unchanged.)
 - At present, the company is selling 8,000 lanterns per month. The sales manager is convinced that a 10% reduction in the selling price will result in a 25% increase in the number of lanterns sold each month. Prepare two contribution format income statement, one under present operating conditions, and one as operation would appear after the proposed changes. Show both total and per unit data on your statements.
 - Refer to the data in (iii) above, how many lanterns would have to be sold at the new selling price to yield a minimum net operating income of \$72,000 per month?

Or,

- ✓ Convoy Manufacturing Company sells a single product. The company's sales & expenses for a recent month as follows: CO3 C3, 10
C4

Particular	Total	Per Unit
Sales	\$ 600,000	\$ 40
Less: Variable expenses	\$ 420,000	\$ 28
Contribution margin	\$ 180,000	\$ 12
Less : fixed expense	\$ 150,000	
Net operating income	\$ 30,000	

Required:

- What is the monthly break -event point in units sold & in sales dollars?
 - Without resorting to computation, what is the total contribution margin at the break-even point?
 - How many units would have to be sold each month to earn a minimum target profit of \$18,000 & \$19,500?
 - Refer to the original data. Compute the companies margin of safety in both Dollar & percentage terms.
5. Rean Devices Ltd. manufactures a smart wearable device. The device requires a special sensor that costs \$4 each unit and is purchased from a local supplier. Rean Devices has prepared a production budget for the device by quarters for Year 2 and for the first quarter of Year 3, as shown below: CO3 C4 10

	Year 2	Year 3

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter
Budgeted Production (Units)	70,000	1,00,000	1,30,000	1,20,000	90,000

The sensor used in the wearable is occasionally subject to supply delays, so the company maintains a buffer inventory. The ending inventory of sensors each quarter must equal 25% of the next quarter's production. At the beginning of Q1 of year 2, 18,000 sensors are in inventory.

Required:

1. Prepare a material purchases budget for the sensors by quarter and for the year in total. At the bottom of your budget, show the dollar amount of purchases for each quarter and the total for the year.
2. Purchase of raw materials is paid in following patterns; 70% paid in the quarter the purchase is made, and the remaining 30% in the following quarter. On January 1, year 2, the company's balance sheet shows 95,500 in accounts payable for raw materials purchase, all of which will be paid on the first quarter of the year.
Prepare a schedule of expected cash payments for material purchase



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester End Examination

Program: B. Sc. in CSE	Semester: Spring-2026
Course Code: MATH-2407	Course Title: Mathematics-IV
Time: 2 hours and 30 minutes	Total Marks: 50

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Separate answer script must be used for separate group.
(iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1	Demonstrate the understanding of the basic principles and operations set theory, complex numbers, geometrical interpretation complex functions and the concept of transformation in a complex plane. different types of functions and signals
CLO2	Apply the concept of transformation of an object into complex space and operation of complex functions
CLO3	Use Fourier series, Laplace's Transforms, Fourier Transform in different scenario.
CLO4	Analyze the harmonics & spectrum of different types of waves.
CLO5	Demonstrate the harmonic analysis using MATLAB.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

1. If $y = f(t) = -1 ; -\pi < t < 0$
 $= 1 ; 0 < t < \pi$
 $f(t) = f(t + 2\pi)$ Here, $T = 2L = 2\pi \therefore L = \pi$
- a) Sketch the function for 3 cycles:
b) Find the Fourier series for the function
- Or
Find the complex form of the Fourier Series expansion of the periodic function $f(t)$ is given by:
- $f(t) = \cos \frac{1}{2} t \quad ; -\pi < t < \pi [T = 2L = 2\pi]$
2. a) Evaluate the convolution sums of $y[n] = x[n] * h[n]$
Where,
 $x[n] = \begin{cases} -1, & n = 0 \\ 1, & n = 1 \end{cases}$ and $h[n] = \begin{cases} 1, & n = 0 \\ 1, & n = 1 \end{cases}$
Where, n represents the time index.
- b) How would you explain the functioning of the cochlea inside the ear as a practical example of a Fourier series?

Mark **CLO** **DL**
2+8 CLO3 App

10 **CLO3** **App**

6 **CLO4** **U**

2 **CLO4** **U**

- c) Plot the line spectrum for the following complex wave

$$x(t) = \underbrace{12}_{\text{Complex wave}} + \underbrace{9\cos(2\pi \times 10t + \frac{\pi}{3}) + 6\cos(2\pi \times 20t - \frac{\pi}{6}) + 4\cos(2\pi \times 30t + \frac{\pi}{4})}_{\text{AC value}}$$

DC value

Group-B

Mark 5 CLO3 DL App

3. a) Find Fourier Transform of

$$\begin{aligned} f(t) &= 1 & ; 0 \leq t < 1 \\ &= -1 & ; -1 \leq t < 0 \\ &= 0 & ; |t| > 1 \end{aligned}$$

Or

Prove that $L(f''(t)) = s^2 L\{f(t)\} - s f(0) - f'(0)$

5 CLO3 App

- b) How many and what are the stages of Convolution Integral?
 c) How to convert an unstable signal into a stable signal? Give an example of an unstable signal (function)
 d) Draw the impulse function $3 * \delta(t - 3)$

1 CLO3 U

2 CLO3 U

2 CLO1 U

4. a) Solve the following Initial Value Problem (IVP) by Laplace Transform
 $Y'' - 3Y' + 2Y = 4e^{2t}$ $Y(0) = -3$ $Y'(0) = 5$

5 CLO3 App

Or

Evaluate $\mathcal{L}\{t^2 \sin 3t\}$ using Multiplication theorem.

5 CLO3 App

5 CLO1 U

- b) Sketch the following function
 $x(t) = -u(t + 3) + 2u(t + 1) - 2u(t - 1) + u(t - 3)$

5. a) Write down a user defined function in MATLAB to reconstruct $f(t)$ in the time interval of $[-4, 20]$ for the following complex wave and also draw line Spectrum for the following wave in one window

8 CLO5 An

$$f(t) = \frac{\pi}{3} + \sum_{n=1}^{\infty} \frac{2}{n} \cos n\pi t$$

2 CLO5 An

- b) If $x[n] = 2$; $n = 0$ and $h[n] = 5$; $n = 0$
 $= 5$; $n = 1$ $= 2$; $n = 1$

Write MATLAB code to find the convolution sum of the above signals.

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Spring 2026

Course Code: CSE-2427 Theory of Computation

Total marks: 50

Time: 2 hours 30 minutes

[Figures in the right-hand margin indicate full marks]

Course Outcomes and Bloom's taxonomy levels are mentioned in additional columns]

Group-A		CO	DL	
1.a)	Show how you can use context free grammar (CFG) to perform the regular operations <i>union</i> and <i>star</i>. OR ✓ Consider the following grammar and show a leftmost derivation and a parse tree for the string aabb. Determine if the grammar is ambiguous using the parse tree. $S \rightarrow AB \mid C$ $A \rightarrow aA \mid a$ $B \rightarrow aB \mid b$ $C \rightarrow aCb \mid ab$	CO1	C4	4
b)	Give the formal definition of Context-Free Grammar (CFG) and Chomsky Normal Form (CNF). Begin with the grammar: $S \rightarrow aAa \mid bBb \mid \epsilon$ $A \rightarrow C \mid a$ $B \rightarrow C \mid b$ $C \rightarrow CDE \mid \epsilon$ $D \rightarrow A \mid B \mid ab$ i) Are there any useless (non-generating) symbols? Eliminate them if so. ii) Eliminate the ϵ-productions. iii) Eliminate the unit productions. iv) Convert the grammar into Chomsky Normal Form (CNF)	CO1	C4	6
	OR ✓ Convert the following grammar in an equivalent grammar in Chomsky Normal Form. $S \rightarrow ABA$ $A \rightarrow B \mid a$ $B \rightarrow b \mid \epsilon$			
2.a)	Construct pushdown automata for the following context free grammar. $A \rightarrow AxB \mid B$ $B \rightarrow ByC \mid C$ $C \rightarrow pAq \mid z$	CO2	C3	3
b)	Using the pumping lemma show that the language $L = \{X^n Y^n Z^n\}$ is not context free.	CO2	C4	3
c)	Design a pushdown automaton that recognizes the following language. $\{a^p b^q c^r d^s \mid p, q, r, s \geq 0 \text{ where } p = s \text{ or } q = r\}$.	CO2	C6	4

Group-B					
3.a)	Give the implementation-level description of Turing Machine that decides the following language: $L = \{ww^R \mid w \text{ is any string of 0's and 1's}\}$ OR $L = \{a^n b^n c^{2n} \mid n \geq 1\}$	C02	C6	3	
b)	Consider a Turing Machine with the transition function δ defined as: $\delta(q_0, 0) = (q_1, 1, R)$ $\delta(q_1, 1) = (q_2, 0, R)$ $\delta(q_2, 0) = (q_2, 1, R)$ $\delta(q_2, _) = (q_{\text{accept}}, _, L)$ The symbol $_$ means blank. Give the sequence of configurations that the machine enters when started with the input string 0100.	C02	C3	3	
c)	Give formal description of a deterministic Turing machine (TM) with complete transition table or state transition diagram that accepts the language $L = \{a^n b^n c^n \mid n > 0\}$ over the alphabet $\Sigma = \{a, b, c\}$.	C02	C6	4	
4.a)	Define (i) computable function, (ii) mapping reducibility, (iii) polynomial time reducibility.	C03	C1	3	
b)	Prove that the set of real numbers is uncountable.	C03	C4	3	
c)	Let the language $A_{\text{NFA}} = \{\langle B, w \rangle \mid B \text{ is an NFA that accepts the input string } w\}$. Prove that A_{NFA} is a decidable language. OR Let $\text{NEQ}_{\text{DFA}} = \{ \langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) \neq L(B) \}$. Prove that "$\text{NEQ}_{\text{DFA}}$ is a decidable language."	C03	C6	4	
5.a)	How can you show that a problem is in class P? Show that the problem of locating an element in a sorted list is in class P.	C03	C4	3	
b)	Let, $\text{HAMPATH} = \{ \langle G, s, t \rangle \mid G \text{ is a directed graph with a Hamiltonian path from } s \text{ to } t \}$. Prove that HAMPATH problem is in NP by constructing a verifier. OR Prove that HAMPATH problem is in NP by constructing a polynomial time nondeterministic Turing machine.	C03	C4	3	
c)	What are complexity classes P, NP, NP-Hard, and NP-Complete? What is the significance of NP-Complete class in complexity theory?	C03	C2	4	

[Answer *all* the questions. Figures in the right hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	BT	Marks
1.a	Define NOT NULL, UNIQUE, and CHECK constraints. Give one SQL example for each constraint.	3	A	4
1.b	Consider a University Database consisting of the relations Department(Dept_ID, Dept_Name, Budget), Student(Student_ID, Name, CGPA, Dept_ID), and Instructor(Inst_ID, Name, Salary, Dept_ID). Answer the following: (a) Write SQL statements to create the three tables by applying appropriate NOT NULL, UNIQUE, CHECK, and FOREIGN KEY constraints. (b) Write an SQL ASSERTION to ensure that no instructor's salary exceeds the salary of the Head of the corresponding department. Explain the purpose of the assertion. (c) Write a TRIGGER that prevents inserting a student whose CGPA is greater than 4.00.	3	A	6
OR				
1.b	Consider the following relation: Student(Student_ID, Name, Department, CGPA) A university wants to secure its Student Information System. Answer the following: (a) Write SQL GRANT statements so that: i. Faculty can SELECT student records only. ii. Department Chairs can SELECT, INSERT, and UPDATE student records. iii. Only the DBA can DELETE student records. (b) Explain how Authentication differs from Authorization. Give one practical example of each in a DBMS. (c) Compare Symmetric-key Encryption and Public-key Encryption. State one advantage and one suitable application of each technique in database security	3	A	6
2.a	Consider the relation schema: EMP(EmpID, DeptID, DeptName, ManagerID, ManagerName, ProjectID) with the following functional dependencies: EmpID → DeptID, ProjectID; DeptID → DeptName, ManagerID; ManagerID → ManagerName Requirements I) Identify all partial and transitive dependencies. II) Determine the highest normal form satisfied by the relation. III) Decompose the relation into Third Normal Form (3NF).	2	A	6
2.b	Define functional dependency and distinguish between trivial and non-trivial functional dependencies with examples.	2	A	4
OR				
2.b	Explain how Armstrong's Axioms can be used to derive new functional dependencies. Illustrate your answer using any suitable relation.			

Group B

Q. No.	Question	CL O	BT	Mark s
3.a	Differentiate between primary indexing and secondary indexing. Which one is more suitable for a file that is frequently searched using non-key attributes?	2	A	4
OR				
3.a	Explain why dense indexing generally requires more storage than sparse indexing. Under what circumstances is sparse indexing preferred?			
3.b	Consider an initially empty B+ Tree where $n=3$.	2	A	6

Insert the following student names one by one:

Rahim, Karim, Salma, Nabila, Hasan, Jamil, Farhan, Tanvir, Rafi, Arif

After completing all insertions:

1. Draw the final B+ Tree.
2. Delete the names Nabila and Salma.
3. Show the B+ Tree after each deletion.

4.a

T1	T2	T3
R(A)		
	R(B)	
W(A)		
	R(A)	
	W(B)	
		R(B)
		W(B)
R(B)		
W(B)		

4 A 4

Consider the above schedule S involving three transactions :

(a)

Construct the

Consider the above schedule S involving three transactions :
(a) Construct the

precedence graph for the schedule.

(b) Determine whether the schedule is conflict serializable. If yes, write the equivalent serial schedule; otherwise, explain why it is not conflict serializable.

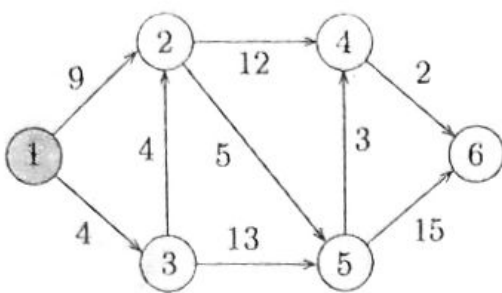
4.b	A banking transaction transfers ₹5,000 from Account A to Account B. (a) Write the sequence of read, write, and arithmetic operations required to perform this transaction. (b) Describe the four ACID properties and explain how each property contributes to maintaining database correctness during the transaction.	4	A	6
-----	--	---	---	---

OR

4.b	Differentiate between Basic Two-Phase Locking (2PL) and Strict Two-Phase Locking (Strict 2PL) with respect to lock releasing, recoverability, and deadlock. Which protocol would you recommend for an online banking system? Justify your answer.	4	A	6
5.a	Differentiate between log-based recovery and check pointing with example.	2	U	4
5.b	Explain the differences among system failure, media failure, and transaction failure. For each type of failure, state the appropriate recovery technique used by a DBMS.	2	U	3
5.c	Explain any three advantages of using a Distributed Database System over a centralized database system.	2	U	3

International Islamic University Chittagong Department of Computer Science and Engineering <i>B. Sc. in CSE Final Examination, Spring 2026</i> Course Code: CSE 2421 Course Title: Computer Algorithms Total marks: 50 Time: 2 hours 30 minutes						
[The figures in the right-hand margin indicate full marks. Course Outcomes and Bloom's Taxonomy Levels are mentioned in additional columns]						
Group A						
				CO	DL	
1a)	Consider an $m \times n$ grid in which a robot starts from cell top-left cell, designated as (1,1), and can move only one step right or one step down until it reaches the bottom-right cell, designated as (m,n). Cost of going right and down is 1 and 2, respectively. (a) Write the recurrence relation of the cost of going to (i,j) location from (1,1). (b) Using bottom-up dynamic programming, compute the minimum cost to go to location (4,5) from (1,1). Show the filled DP table as it is built. (c) List the cells visited on one optimal (minimum-cost) path.			CO 4	C4	5
1b)	Write the DP recurrence $m[i][j]$ for the minimum number of scalar multiplications Matrix Chain Multiplication problem and identify the optimal substructure it relies on.			CO 4	C4	3
1c)	Given an array of n integers, the Longest Decreasing Subsequence (LDS) is the longest subsequence whose elements appear in strictly decreasing order. Define the DP state and write the $O(n^2)$ recurrence relation for the length of the LDS.			CO 4	C4	2
2a)	Explain the difference between worst-case complexity and amortized complexity using dynamic array insertion as an example. OR Why is insertion at the end of a dynamic array considered amortized ($O(1)$) , although resizing may take ($O(n)$) time? Show the amortized cost calculation.			CO 2	C5	3
2b)	You're given a different version of vector push back operation – one that adds a constant number of elements to the array's size every time it reallocated, rather than multiplying it – What would be the amortized running time of this version? OR Analyze the process of splitting a full B-tree node during insertion. Evaluate the effects of promoting the smallest, middle, and largest keys, and justify why promoting the middle key is the most appropriate choice for preserving B-tree properties.			CO 4	C4	3

2c)	Consider a B-tree of minimum degree $t=3$. Insert the following keys into the empty B-tree in the given order: 10, 20, 30, 40, 50, 60, 70 After constructing the initial B-tree, perform the following deletions one by one in this exact order: 70, 10, 20 Show the B-tree after each deletion and explicitly state which of the following deletion cases occurred in that step: 1. Deletion from a leaf node without underflow 2. Deletion using borrowing from a sibling 3. Deletion using merging with a sibling OR Insert the keys 10, 20, 5, 6, 12, 30, 7, 17, 3, 25, 40 into a B-tree of minimum degree ($t = 2$), and identify at which insertions node splitting occurs.	CO 2		
Group B				
3a)	You are given a directed graph G1 with vertex set $V = \{A, B, C, D, E, F\}$ and edge set $E = \{(A,B), (A,C), (B,D), (C,D), (C,E), (D,F), (E,F), (F,A)\}$. Write how the D value (distance from the source) and the queue content is changed for each vertex after each iteration of a breadth first search with vertex A as the source. An iteration starts when a vertex is extracted from the queue. Before the first iteration, D value and queue content are as follows. $D = [0, \text{INF}, \text{INF}, \text{INF}, \text{INF}, \text{INF}]$ $Q = [A]$	CO 3	C3	3
3b)	In what order will the vertices of graph G1 of question 3a be visited in a depth first search? Can you find a topological sort of the graph? Justify your answer.	CO 3	C3	3
3c)	Consider the undirected, weighted graph $G = (V, E)$ where $V = \{1, 2, 3, 4, 5, 6\}$. The edge list is: (1, 3, 2), (3, 5, 3), (1, 2, 4), (2, 5, 5), (4, 5, 6), (5, 6, 9), (2, 4, 10), (4, 6, 11). The edge list is given (u, v, w) format where u, v are the endpoints and w is the weight of edge (u, v). Find the minimum spanning tree of G using Kruskal's algorithm and show each step. OR Find the minimum spanning tree of G using Prim's algorithm and show each step.	CO 3	C3	4
4a)	What is path relaxation? Briefly describe with necessary figures.	CO 3	C4	2
4b)	The Bellman-Ford algorithm runs a relaxation step $V-1$ times. (a) Explain and show why $V-1$ iterations are sufficient to guarantee the shortest paths are found if no negative-weight cycles exist. (b) Then Explain why executing a V-th iteration allows us to identify the presence of a negative-weight cycle reachable from the source.	CO 3	C4	4

4c)	 Using vertex 1 as the source, apply the Dijkstra's algorithm to compute the shortest path distances from 1 to all other vertices. Show the distance values after each iteration of the outer loop is completed. OR Using vertex 1 as the source, apply the Bellman-Ford algorithm to compute the shortest path distances from 1 to all other vertices. Show the distance values after each iteration of the outer loop is completed.	CO 3	C3	4
5a)	Consider a flow network $G = (V, E)$ with source s and sink t. The capacities of the directed edges are given below: $(s, v_1) = 16, (s, v_2) = 13, (v_1, v_2) = 10, (v_1, v_3) = 12, (v_2, v_1) = 4,$ $(v_2, v_4) = 14, (v_3, t) = 20, (v_4, v_3) = 7, (v_4, t) = 4$ Execute the Ford-Fulkerson algorithm step-by-step. For each iteration, state the augmenting path selected, its bottleneck capacity, and update the residual graph network.	CO 3	C3	4
5b)	A backtracking algorithm solves the N-Queens problem by placing one queen in each row and, for every row, trying the columns in increasing order. (i) Two queens occupy rows i and j in columns c_i and c_j, respectively. Derive the algebraic condition on these indices under which the two queens lie on a common diagonal, and explain why placing exactly one queen per row makes any row-conflict check unnecessary. (ii) Describe the structure of the state-space tree the algorithm explores, and explain precisely what distinguishes a node that is abandoned (pruned) from one that is expanded further. (iii) A classmate argues that backtracking and the brute-force method that lists all N^N column assignments share the same worst-case time complexity, so backtracking offers no real advantage. Assess this argument, distinguishing clearly between worst-case asymptotic cost and observed running time on typical inputs.	CO 4	C3	4
5c)	For each of the following problems, determine whether it belongs to P, NP, NP-Complete, or NP-Hard, and justify your answer. • Sorting • Traveling Salesman Decision Problem • Traveling Salesman Optimization Problem • Hamiltonian Cycle OR A problem X satisfies the following conditions: • A proposed solution can be verified in polynomial time. • Every problem in NP can be polynomially reduced to X. Analyze the complexity class of problem X. Justify whether it belongs to P, NP, NP-Complete, or NP-Hard	CO 4	C4	2